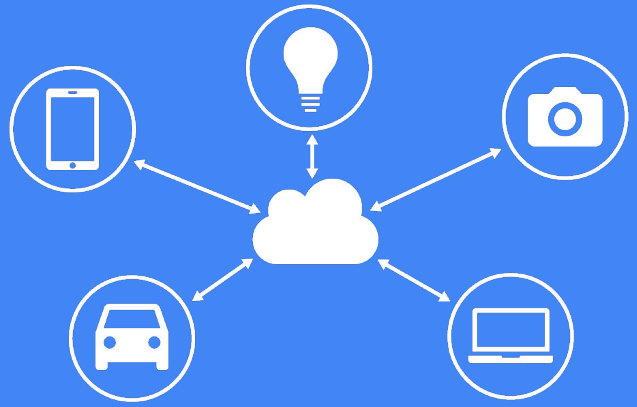
IoT introduction. Technologies overview. Preparing to workshops



by Volodymyr Shevchyk

Indeema

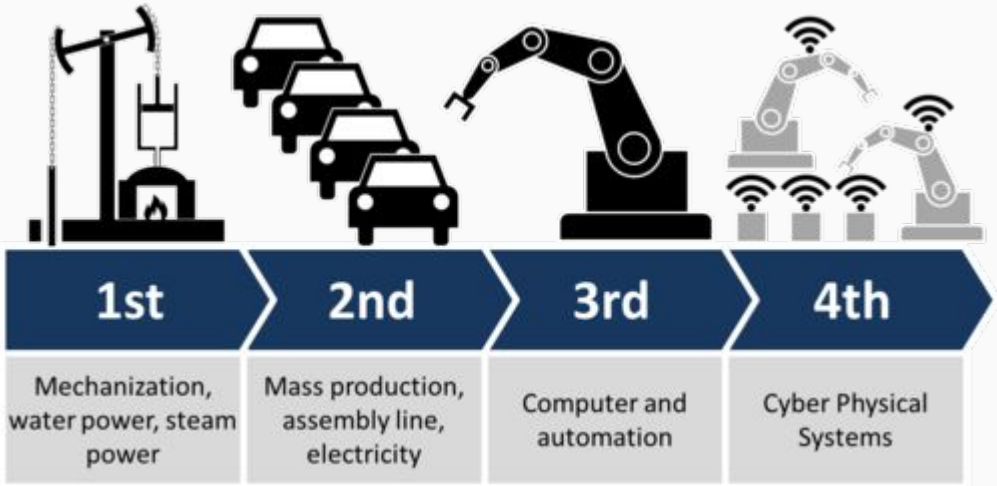
Contents

- What is it all about?
- Some IoT projects
- IoT technologies overview
  - Data acquisition
  - Data transaction
  - Data analysis
- Preparing to workshops

What is it all about?

What is it all about?

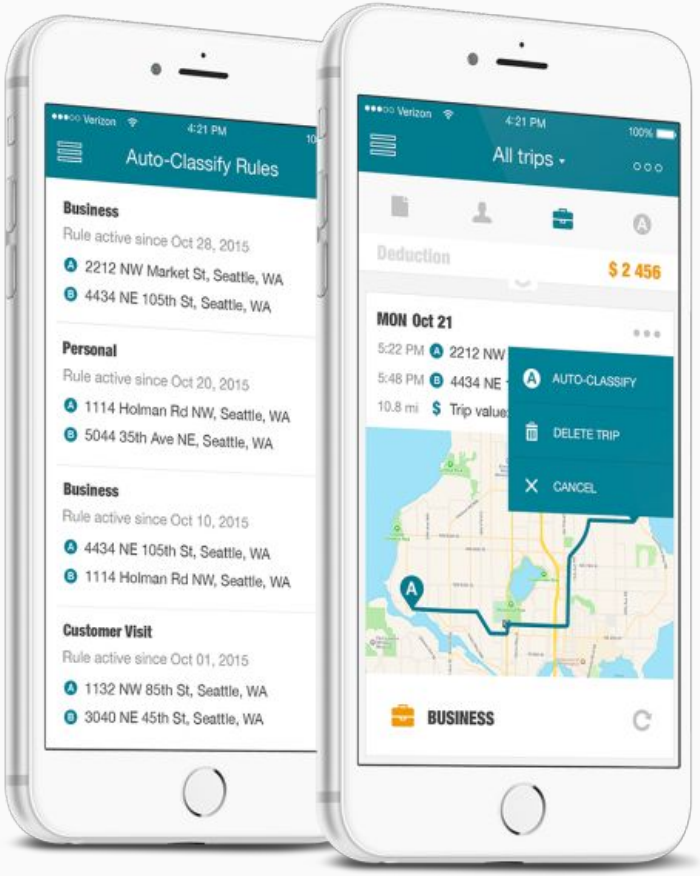
- New conception
- New way of optimisation
- Main part of industrial revolution 4.0
- Possibility to predict and prevent potential disaster



Some IoT projects

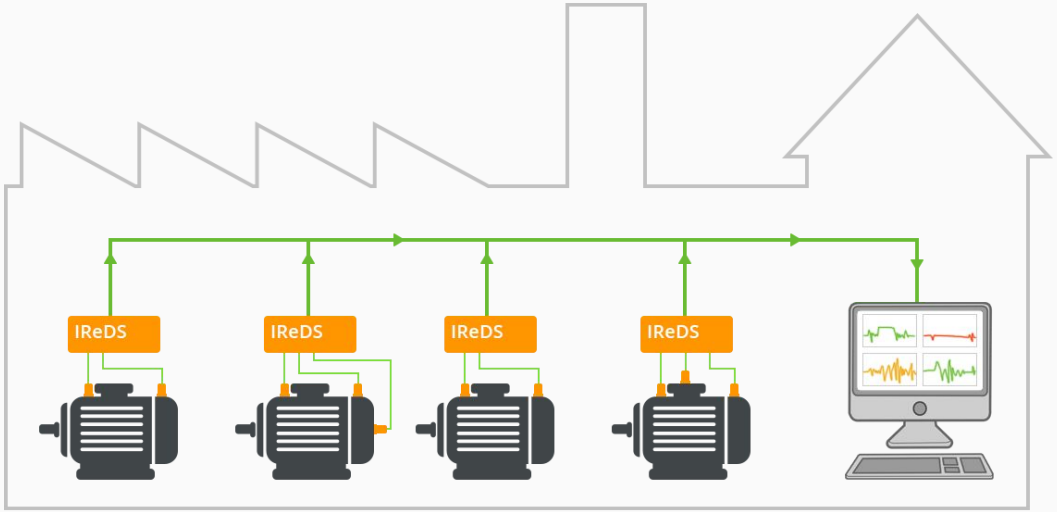
MileCatcher

- Track trips
- Create reports



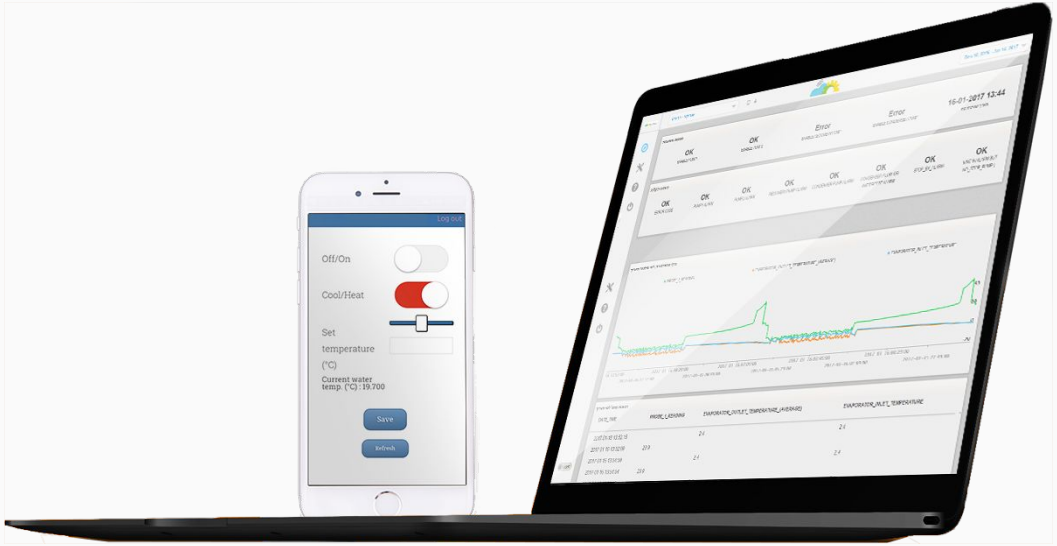
iReDS

- Records vibration level
- Provides predictions
- Provides detailed reports



EcoPlant

- Records data from sensors
- Provides reports



Smart Dress

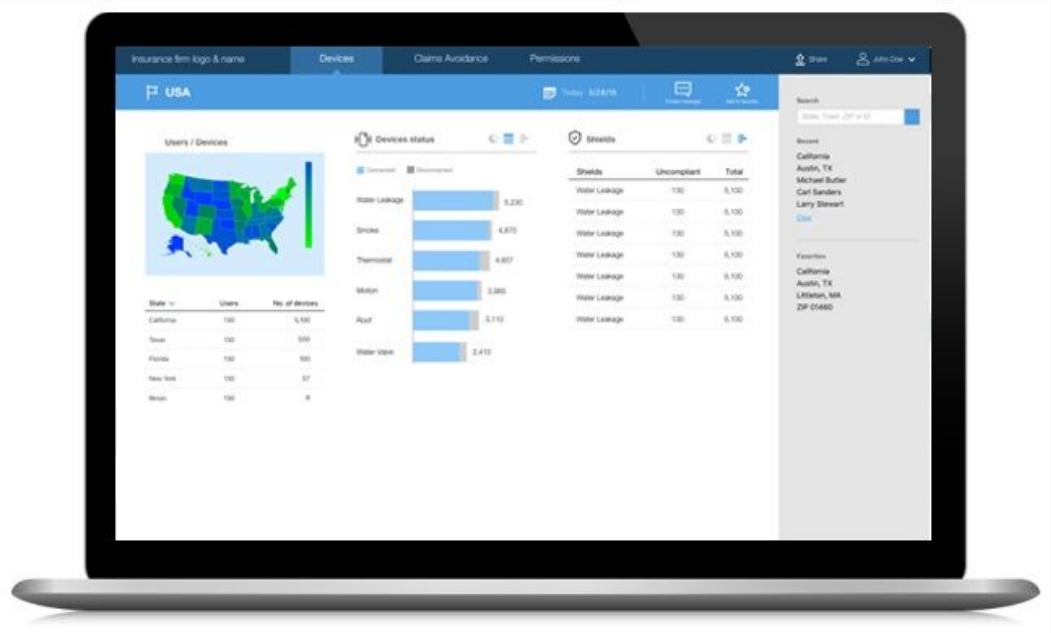
- For sportsman, all people
- Records body signals
- Sync with server
- Provides diagnostics
- Emergency calls

Smart accounting

- Automatically reports

Smart homes, cities

- Level of risk in each region
- Automatic risk detection
- Smart insurance



Nature

- Weather
- Volcanoes
- Oceanes
- Glacier

IoT technologies overview

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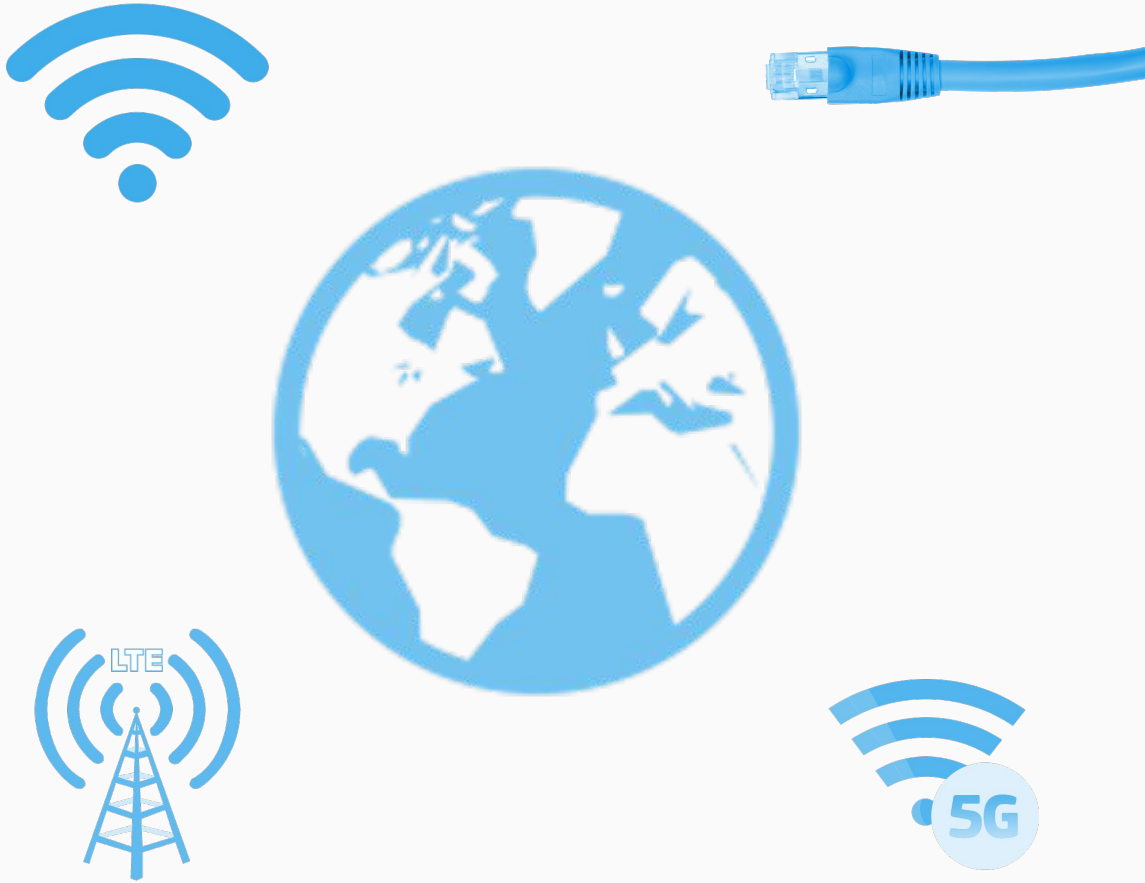

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IoT technologies overview

- Data acquisition
- Data transaction
- Data analysis

amazon web services

Microsoft Azure

IBM Watson

SAMSUNG ARTIK™ Cloud

PREDIX

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Preparing to workshops

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- Arduino starter kit
- Qt 5.9
- Predix cloud
- QPredix SDK

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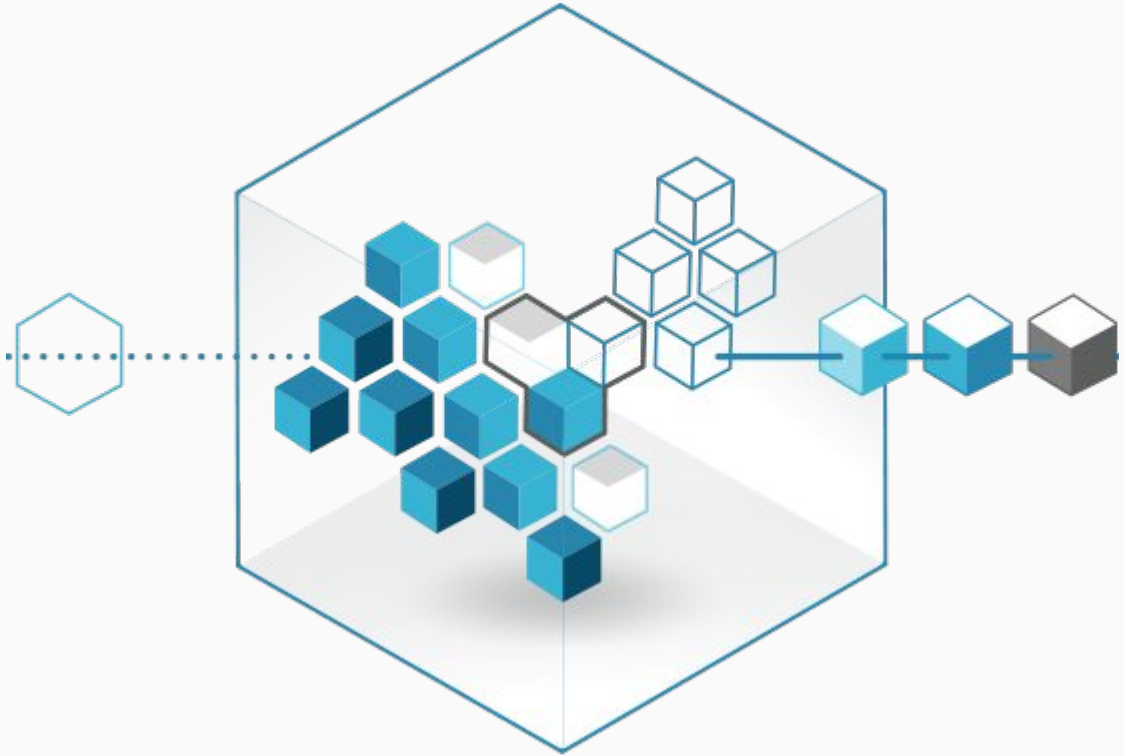

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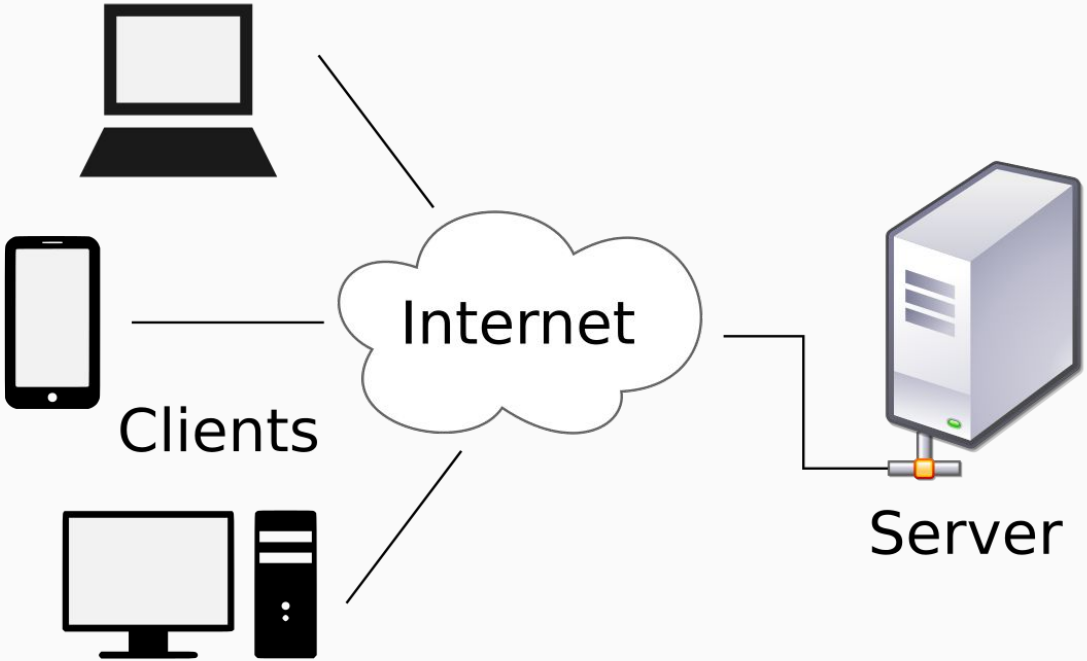

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Preparing to workshops

- Register at https://www.predix.io/registration/
- Install Qt 5.9 from https://www.qt.io/download-qt-for-application-development
- Download QPredix SDK - https://github.com/IndeemaSoftware/QPredix
- Install Arduino Studio from https://www.arduino.cc/en/Main/Software
- Prepare Arduino starter kit with temperature sensor, prototyping board and Arduino uno
- Learn little C/C++, Qt
- Register at https://goo.gl/YvWPWj

Thank you!